

1.

The matrix_task is performing very computationally intensive operations:

- It executes a **10x10 matrix multiplication**.
- It utilizes 4 nested loops (i, j, k, l).
- It includes a simulation delay of **1 billion iterations**.
- Consequently, it occupies the CPU for an extended period.

2.

In FreeRTOS, the highest priority task always executes first.

- Initially, matrix_task is **Priority 3** (High) and communication_task is **Priority 2** (Low).
- When the Matrix task hogs the CPU, the Communication task is forced to wait (starvation).
- Although the Communication task is scheduled to run every 200ms, it cannot get CPU time until the matrix processing finishes.
- When the priority of the Communication task is increased (to **Priority 4**), it is able to run before the Matrix task.

3.

The completion time of matrix_task increases.

Reason:

- When the Communication task is set to Priority 4, it **preempts** the matrix_task.
- The Communication task interrupts the Matrix task while it is running.
- The Matrix task pauses, waits for the Communication task to finish, and then resumes.
- Due to these interruptions and context switches, the total execution time of the matrix task is prolonged.

4.

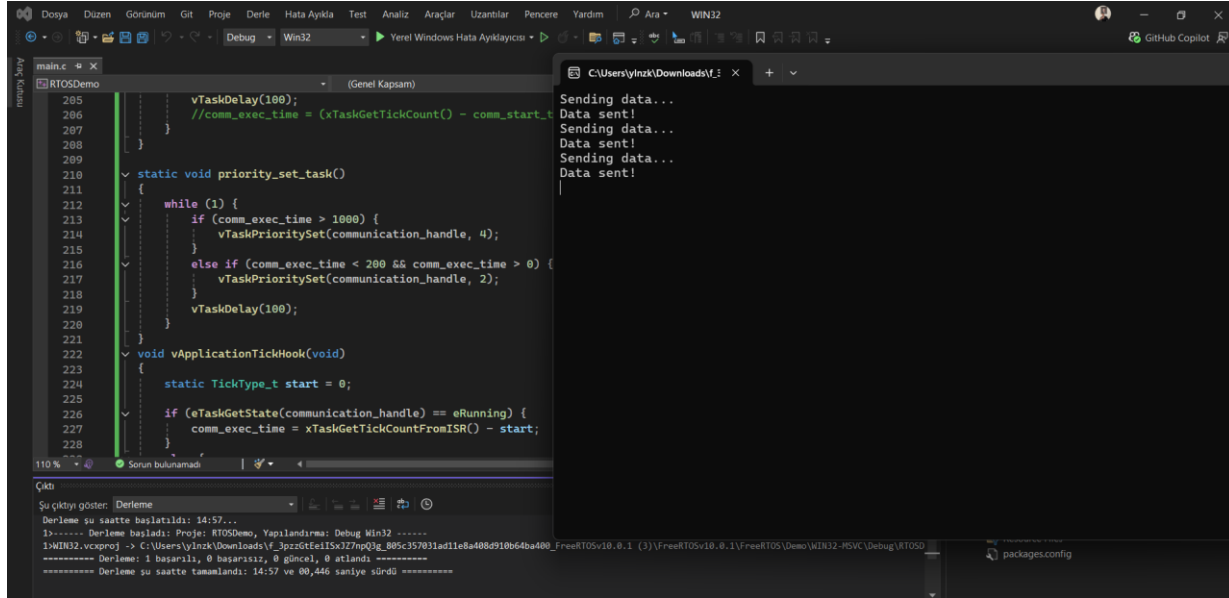
Period = Matrix Execution Time + 100ms

To measure the exact duration, you can use vApplicationTickHook():

1. Record the start tick when the Matrix task begins.

2. Record the end tick when it finishes.
3. The difference is the Period (convert ticks to seconds).

Estimation: Considering the simulation delay plus the matrix multiplication, it will likely take **several seconds** (approximately **3–5 seconds**, depending on the system clock speed).



```
main.c
205 vTaskDelay(100);
206 //comm_exec_time = (xTaskGetTickCount() - comm_start_t
207 }
208
209 static void priority_set_task()
210 {
211 while (1) {
212 if (comm_exec_time > 1000) {
213 vTaskPrioritySet(communication_handle, 4);
214 }
215 else if (comm_exec_time < 200 && comm_exec_time > 0) {
216 vTaskPrioritySet(communication_handle, 2);
217 }
218 vTaskDelay(100);
219 }
220
221 void vApplicationTickHook(void)
222 {
223 static TickType_t start = 0;
224
225 if (eTaskGetState(communication_handle) == eRunning) {
226 comm_exec_time = xTaskGetTickCountFromISR() - start;
227 }
228 }
```

```
110 %
Çıktı
Şu çıktıyı göster: Derleme
Derleme şu saatte başlatıldı: 14:57...
1>----- Derleme başladı: Proje: RTOSDemo, Yapılandırma: Debug Win32 -----
1>WIN32.vcxproj -> C:\Users\ylnzk\Downloads\FreeRTOS\10.0.1 (3)\FreeRTOS\10.0.1\FreeRTOS\Demo\WIN32-PSVC\Debug\RTOSD
----- Derleme 1 başarıldı, 0 başarısız; 0 g@ncel, 0 atlandı -----
----- Derleme şu saatte tamamlandı: 14:57 ve 80,446 saniye sürdü -----
```